Detailed review – Team Project 1 (pendulum), task by task

Follows the assignment (`projects/01NAEX_Project1.ipynb`, tasks 1–11 and their bullet points). For each question: what we ask and why, the answer and review of team G, the answer and review of team Bl, and a summary.

- **Team G** = Galloppi, Paleta, Remiš, Veselý (`01NAEX_Project1_Galloppi_Paleta_Remis_Vesely`).
- **Team Bl** = Blažek, Guliev, Linazisoro, Morovicsová (`01NAEX_Project1_Blazek_Guliev_Linazisoro_Morovicsova`).
- "Part A" and "Part B" are parts of the experiment (RCBD and Graeco-Latin squares), not teams.

Numbers not visible in the notebooks (team Bl prints its conclusions via `display(Markdown)`) were recomputed from the teams' CSV files; they are marked "(recomputed)".

One pendulum instead of three: both teams used a single pendulum and changed its length. The instructor allowed this to speed up the measurements, so it is not a mistake in itself. What is assessed is whether the team **acknowledged** that the experiment is therefore not fully randomized, and whether it **recorded the actual run order**, so that time trends and the error structure can be assessed.

Task 1 – Experimental setup

1.1 Apparatus, measurement of L and θ_0, timing and N

Question: describe the pivot, string, rings (mass, size), how length and angle were measured, how timing was done and the number of oscillations N. **Why we ask:** without a precise description we cannot judge systematic errors (length to the centre of mass, the ring correction R^2/L, rigidity of the pivot) or reproduce the experiment. The choice of N directly determines precision: a reaction-time error of 0.05–0.2 s is divided by N.

Team G – answer:

- One pendulum hung from a sponge holder at the whiteboard, string fixed with tape. Steel rings, 1/2/3 pieces, weighed 10.08 / 20.18 / 30.38 g (20.26 g recorded in Part A, the 0.08 g difference is commented).
- Length 0.35 / 0.65 / 1.00 m measured with a tape measure from the edge of the holder to the centre of the rings after every re-tying. Ring correction R^2/L not applied ("below the precision of the tape measure").
- Angle 5°, 30°, 45° set via $d = L \sin\theta_0$; 5° instead of 10° on purpose, to increase the difference.
- $N = 10$ (compromise between time and amplitude decay), phone stopwatch, start at the lowest point after one swing. In 4 runs operator A counted only 9 oscillations; analysed with $N = 9$, noted, nothing deleted.

Review:

- The description is complete and concrete, including reasons for the deviations ($N = 10$, 5°). The ring radius is missing (the R^2/L claim cannot be checked; for $R \approx 1$ cm and $L = 0.35$ m it is 0.03 cm, so the claim is probably true), and there is no photo.
- One pendulum instead of three is an allowed deviation. The team states explicitly that, because of re-tying, the length was not fully randomized and that a time trend may be partly confounded with the length effect. **This is exactly what was expected.** They did not, however, think through the consequences for the error term (Task 3).
- Only the **nominal** length went into the CSV, although the text says it was measured after every setting. Task 7 therefore does not use "measured values".
- 5° at $L = 0.35$ m means $d = 3$ cm; an error of ±0.5 cm is ±1°, irrelevant for the angle effect (0.05 %). Fine.

Follow-up questions: What is the ring radius, and how much would R^2/L change L at 0.35 m? Where are the actually measured lengths?

Team Bl – answer:

- Pendulum hung from a rack made as rigid as possible; the string length was changed between 0.35, 0.65 and 1.00 m ("We changed the length of the string"), i.e. also a single pendulum. Rings 10.14 / 20.22 / 30.48 g.
- L measured from the pivot to the centre of mass of the rings "taking into account the inner radius", but the table says "to the approximate centre of the bob".
- Angles 10°, 30°, 50° from marks on paper.
- $N = 5$ in Part A, $N = 10$ in the pilot and Part B. No reason given.

Review:

- The description is more general: pivot and string, ring size and a photo are missing.
- The single pendulum is allowed, but the team **nowhere acknowledges** that changing the length means the experiment is not fully randomized, nor says how often the length was changed.
- $N = 5$ **in Part A is the main weakness of the whole project.** The assignment recommended $N = 20$. With $N = 5$ a reaction error of 0.1 s is 2 % of the period at 0.35 m, and one miscounted oscillation means −20 %. Consequences: $\hat{\sigma}$ in Part A is 3.3 % of the period (team G 0.9 %), and block 3 contains obvious counting errors (Task 4). The deviation is acknowledged but **not justified**.
- Different N in different parts also made the pilot useless for planning Part A (different procedure).
- 3 rings instead of 2 in the pilot and Part A: acknowledged, irrelevant for the results.

Follow-up questions: Why $N = 5$? What percentage of the period is a 0.1 s error at $N = 5$ and $L = 0.35$ m? How exactly did you measure L ("approximate centre")? How many times did you reset the length?

Summary 1.1: Both teams used the allowed single pendulum. Team G described the apparatus better, justified its deviations and acknowledged the restricted randomization. Team Bl did not acknowledge it and chose $N = 5$, which destroyed the precision of Part A. Both should have recorded the actually measured lengths and the ring radius.

1.2 Nuisance factors and how they are handled (block, randomize, keep constant)

Question: list the expected nuisance factors and say for each whether it is blocked, randomized or kept constant. **Why we ask:** this is the core of L01 (three principles: replication, randomization, blocking) and L03 ("block what you can, randomize what you cannot").

Team G – answer: a table of six factors: operator (block in A, rows in B), time drift (randomization, in B also the square factor), releaser (release from rest at the mark), amplitude decay (constant N), errors in L and θ_0 (measured after every change), pivot movement and air currents (constant).

Review: a model answer, exactly as in L01/L03. Inconsistency: for drift they claim randomization, which they restricted in Part A and did not actually perform in Part B (Task 3). They acknowledge this for Part A only.

Follow-up question: Which factor in your table was in the end not handled by randomization?

Team Bl – answer: operator (block in A, row in B); air, friction at the pivot, string stretching, release technique "minimized by using the same equipment and procedure"; "timing habits, damping and counting errors can affect the results".

Review: the factors are listed, but the assignment "block / randomize / constant" is missing for each of them. **Randomization and time drift are not mentioned at all**, although the pilot shows a strong trend (Task 2.1) and changing the length restricts randomization. Counting errors are named but nothing is done against them.

Follow-up question: How did you protect the experiment against a time trend (learning, fatigue)?

Summary 1.2: Team G answered correctly and completely. Team Bl only listed factors; the principle of randomization is missing.

1.3 Which factors are nuisance factors in Part A and which in Part B

Why we ask: to check that students understand that the same variable can be a nuisance factor in one design and a factor of interest in another (mass).

Team G: in A the operator (block) and the run order; mass constant. In B the operator in rows and the square as a second blocking factor; mass is a factor of interest (because of Task 8). **Correct.**

Team BI: only implicitly (operator as block in A, rows in B). No explicit answer.

Summary 1.3: only team G answered correctly.

Task 2 – Pilot study and sample size

2.1 Pilot data, $\bar{T}$, $\hat{\sigma}$ and comparison with theory

Question: 10 runs by one operator at 0.65 m, 30°, 2 rings; compute $\bar{T}$, $\hat{\sigma}$ and compare with `theory_period()`.
Why we ask: $\hat{\sigma}$ is the input for the power calculation (L01, L03). Comparison with theory is the first check for systematic errors of the apparatus.

Team G – answer: 7 separate runs by operator A + 3 Part A runs at the same setting (operators C, B, A). $\bar{T} = 1.6428$ s, $\hat{\sigma} = 0.00957$ s, theory 1.6455 s (difference 0.16 %).

Review:

- Calculation and comparison correct; agreement with theory is excellent.
- The pilot is **not** "one operator, 10 runs": it mixes 7 consecutive runs of operator A with 3 runs of three different operators. The team says so openly.
- Chronology: runs 8–10 belong to Part A. Formally, the plan $b = 3$ was based on data that were already part of the main experiment. In practice it does not matter ($b = 3$ is the minimum required), but it is worth asking.
- $\hat{\sigma}$ from 10 runs is uncertain: the 95 % CI is (0.0066; 0.0175) s. Team G reports this in Task 11 – correct.

Follow-up question: When exactly did you decide on $b = 3$: before run 8 or after run 10?

Team BI – answer: 10 runs, 3 rings, $N = 10$. $\bar{T} = 1.5827$ s, $\hat{\sigma} = 0.0290$ s, theory 1.6455 s, the mean is 3.8 % below theory (recomputed).

Review:

- Calculation correct. But the pilot used $N = 10$ and Part A $N = 5$, so it does not represent the Part A procedure. The team says so and adds a scenario "SD doubled for $N = 5$". A reasonable reaction, but repeating the pilot would have been better.
- **Undetected trend:** T increases in measurement order from 1.54 to 1.63 s (Spearman 0.87, $p = 0.001$). $\hat{\sigma}$ is therefore inflated by drift, and the −3.8 % deviation from theory is partly operator learning. The team does not comment on the deviation from theory at all.
- The note "Corrected pilot" in the data is not explained.

Follow-up questions: Plot the pilot in measurement order. What do you see? What does "Corrected pilot" mean? Why is the pilot 3.8 % below theory?

Summary 2.1: Team G has a good pilot (with the mixture of sources acknowledged). Team BI's pilot has a different N, a trend and a large deviation from theory, none of which was analysed.

2.2 Power of the F-test for the angle as a function of the number of blocks b

Question: target = theoretical difference 10° vs 30° at the middle length, least favourable configuration, noncentral F; choose $b \geq 3$. What to do if the power is too low? **Why we ask:** planning the size of an experiment from a pilot is a key skill from L01 and L03 ($\lambda = nD^2/(2\sigma^2)$, $n = 3b$ per level, $df_E = 8(b-1)$).

Team G – answer: $\lambda = 3bD^2/(2\hat{\sigma}^2)$, $df_1 = 2$, $df_2 = 8(b-1)$, correctly derived. Power 92 % for $b = 2$, 99.7 % for $b = 3$. They chose $b = 3$. To increase power: a larger N reduces the reaction error but increases amplitude decay.

Review:

- Methodologically exactly as in L03.
- Power is computed with the point estimate $\hat{\sigma}$. With the upper CI limit (0.0175 s), $b = 3$ would give only 70 % and $b = 2$ 44 % (recomputed). Planning with the upper limit or with a margin is good practice; the team did not mention it.
- The calculation is on the T scale in seconds and for one length. On a shorter pendulum the angle effect is smaller in seconds, so 0.65 m is not the worst case on the T scale. On the $\log T$ scale the effect is the same for all lengths.

Follow-up question: What would the power be if the true σ were at the upper limit of the pilot interval?

Team Bl – answer: the same (correct) formula. Power table for $b = 2$ to 200: 60 % for $b = 6$, 80 % only at $b = 9$, under 20 % in the "$N = 5$" scenario. Explicitly: "It was made after measurement, so it is not evidence of an earlier plan."

Review:

- Formula and interpretation of power are correct; the explanation of what 80 % power does and does not mean is very good.
- **But the task was not done:** $b = 6$ was not derived from the pilot and, by their own calculation, is underpowered. The actual power for the 10° vs 30° target was 21 % (Task 11).
- On "what else could you change" they answer well (longer timing, better equipment, wider angle range), but did the opposite themselves (shorter timing).

Follow-up question: How did you choose 6 blocks?

Summary 2.2: Both use the correct formula. Only team G planned as required. Team Bl honestly admitted that the calculation is post hoc.

2.3 Error degrees of freedom of one and two Graeco-Latin squares

Why we ask: to understand the saturation of a 3×3 square (L04): 9 observations, 1 + 4·2 = 9 parameters.

Team G: one square 0 df ($8 - 4 \cdot 2 = 0$), two squares 6 df ($17 - 1 - 4 - 3 \cdot 2$), with a breakdown by factor. **Correct.**

Team Bl: 9 parameters for 9 observations → 0 df; two squares 12 parameters for 18 → 6 df, assuming the same effects in both squares and no interactions. **Correct**, with the assumptions stated.

Summary 2.3: both correct.

2.4 Final plan before the main experiment

Why we ask: the plan should be fixed before the data, otherwise the analysis may be adapted to the results.

Team G: $L \in \{0.35, 0.65, 1.00\}$, $\theta_0 \in \{5^\circ, 30^\circ, 45^\circ\}$, $b = 3$, $N = 10$, roles. **Done.** (No mention of the single pendulum and the order of lengths, which later turned out to be the main limitation.)

Team Bl: no plan before the measurements; Section 3 explicitly generates a plan "for a repeat experiment". **Not done**, acknowledged.

Summary 2.4: only team G.

Task 3 – Randomized run sheets

3.1 Part A: random order within each block, with a fixed seed

Why we ask: a block in an RCBD must contain all treatments in random order (L03). A fixed seed makes the plan verifiable.

Team G – answer: no seed. They show the actual order: first all runs at 0.65 m, then 0.35 m, then 1.00 m; within a length the angles almost always ascending; the three operators measured the same setting one after another, only the order of operators was drawn at random. The deviation is acknowledged in Task 1 (Limitations).

Review:

- Acknowledged and justified (re-tying takes time), and the actual order is recorded in the `run` column. With the allowed single pendulum this is **the required minimum, and the team met it**.
- Grouping by length was necessary, but three consecutive runs of the same combination were not. The order of angles and operators within a length could have been fully randomized.
- Consequences not thought through:
  - The time trend is confounded with length and, within a length, with angle.
  - The length was set **once** per level. The three "replicates" share the same tying and the same measurement of L. The effect of L is therefore unreplicated, and the residual contains only the timing and release error, not the error of setting the length. This is a split-plot structure (L11): the CIs for β_1 and g are probably too narrow. Consistent with this, g from Part A differs by length (10.04 / 9.86 / 9.88) more than the interval suggests.
  - Block = operator, but operators alternate run by run. A "block" is therefore not a contiguous session; blocking can capture nothing but the personal timing style.

Follow-up questions: Are three consecutive runs of the same combination independent replicates? What does the residual measure? How many independent length settings do you have?

Team Bl – answer: a plan with seed `20261026` for 6 blocks, but described as "an example plan for a **repeat experiment**". Actual order not recorded (`run` empty; `source_order` is the row order of the paper protocol: within a block L descending, angle ascending).

Review:

- Task not done. A plan generated after the measurements proves nothing.
- The team changed the length of one pendulum, so the measurements were not fully randomized, but it **does not acknowledge this** and **did not record** the actual order. These are exactly the two things that were mandatory with the allowed single pendulum.
- The protocol row order (L descending, angle ascending within a block) suggests that the length was reset in each block. If so, the length effect is replicated across blocks (split-plot with length as the whole-plot factor), which is better than in team G. Without the recorded order this cannot be verified.
- Independence of runs and protection against a time trend therefore cannot be assessed. The team does not raise the question at all.

Follow-up questions: In what order did you actually measure? How many times did you reset the length? Was the order of angles within a block random?

Summary 3.1: The single pendulum was allowed for both teams. Team G acknowledged that the experiment is not fully randomized and recorded the actual order; it met the minimum, although it did not think through the consequences for the error. Team Bl neither acknowledged nor recorded it. Correct approach: randomize the order of lengths in each block and the order of angles within a length (`rng.permutation`), acknowledge the restriction and record the order in `run`.

3.2 Part B: two independently randomized squares, `random_graeco_latin_3()`

Why we ask: randomizing rows, columns, symbols and run order protects the square against systematic error (L04).

Team G – answer: seeds `31273448` and `70415791`; check "SUCCESS: The generated design squares match the recorded Part B data perfectly!" and matching run order.

Review (serious):

- Sorted by `order`, the runs in **both** squares are exactly lexicographic: 1, 1, 1, 2, 2, 2, 3, 3, 3 rings and within that length 0.35, 0.65, 1.00. The probability of such an order is $1/9! \approx 2.8 \cdot 10^{-6}$ per square; among 20 000 trial seeds it never occurred. The seeds were almost certainly searched for so that the "random" plan matched a convenient (or already executed) order.
- Consequence: mass is **completely confounded with position in time** in both squares (1 ring = runs 1–3, 3 rings = runs 7–9). The mass effect estimate +1.0 % (Tasks 6 and 8) cannot be distinguished from drift within the square.
- The "SUCCESS" check only verifies that the data match the plan, not that the plan is random.

Follow-up questions: How did you choose the seeds? What is the probability that a random order follows exactly the number of rings and the length? What does that mean for the mass effect?

Team Bl – answer: two squares with seeds `20261027`, `20261028`, again "for a repeat". The actual squares in the data are valid (orthogonality checked in Task 4) but differ from the generated ones.

Review: task not done. We do not know whether the actual square was randomized or in what order it was measured. In Part B the length changes in every run, so a random order is laborious with a single pendulum; all the more reason to state how the measurements were actually done.

Follow-up question: How did you choose the actual layout of both squares?

Summary 3.2: neither team has a documented randomization of Part B. For team G, moreover, randomization is claimed that did not actually take place. Correct approach: generate the plan **before** the measurements with any seed and measure in that order, even if inconvenient.

3.3 Actual order in the `run` column

Team G: `run` 1–52 filled in. **Done.** `date_time` empty. **Team Bl:** `run` and `date_time` empty. **Not done. Summary 3.3:** only team G. Without `run`, the diagnostics in Task 9 are impossible.

Task 4 – Data and descriptive analysis

4.1 Data check (runs per cell, block, square, missing values, notes)

Why we ask: data errors must be found before modelling, not after.

Team G: `info()`, counts of notes, cross-tables of runs per cell × block and square × operator × rings. Commented: only `date_time` missing, 7 + 3 pilot runs, 4 runs with $N = 9$. **Correct and complete.**

Review G: all 4 runs with $N = 9$ belong to operator A in Part B. Two of them (2 rings, 0.35 m) are 1.9 % and 2.4 % above theory, more than the other runs (recomputed). The adjustment of N may therefore be inaccurate and affects the operator effect. **Follow-up question G:** How did you find out that the operator counted only 9 oscillations?

Team Bl: automatic `assert` checks (counts, positive values, block balance, orthogonality of both squares), table of missing metadata. "All 82 supplied readings are retained, including the low A readings for group C in repetition 1."

Review Bl:

- The structural checks are excellent (the most systematic of the two teams).
- They noticed the low values in block 3 but **did not investigate**. For example $L = 0.65$ m, 50°: $t_5 = 6.98$ s corresponds to 4.1 actual periods, almost certainly a counting error. The ratio T/theory in block 3 is 0.917 on average (min 0.822). Correct approach: check the protocol, mark in `note`, analyse with and without these points.

Follow-up question Bl: How many periods correspond to $t_5 = 6.98$ s at 0.65 m? What should be done with such a point?

Summary 4.1: Team Bl has better structural checks, team G better handling of notes. Neither investigated suspicious runs in depth.

4.2 Tables of means and SDs (T by L and θ_0 in A, by mass in B)

Team G: both tables. Comment: 5° and 30° are close (small angles), 45° clearly higher; SD by mass ≈ 0.37 s because of the lengths. **Correct.** The sentence "with increasing number of rings the pendulum gets slower" is speculation; the differences are not checked.

Team Bl: both tables (with counts). Comment only on the SD by mass (contains the length effect). **They do not comment on the huge SDs in Part A** (0.03–0.12 s; 0.125 s at 0.65 m and 50° because of block 3). This was already a sign that something was wrong with the data.

Summary 4.2: both done. Team Bl missed the signal in the SDs.

4.3 Plots: interaction plot on T and $\log T$, T by operator, T (or deviation from theory) in run order

Why we ask: the interaction plot shows why the $\log T$ scale is additive; the plot in run order is the only check for a trend.

Team G: all four plots (interaction plot of A + B together, boxplot by operator, T vs `run`). Comment: "parallel lines … no interaction on either scale"; no difference between operators; the trend check is moved to residuals in Task 9 because T is dominated by length.

Review G: complete. The claim "parallel on both scales" contradicts theory and their own Task 5a: on the T scale the angle effect grows with length (0.047 s at 0.35 m, 0.079 s at 1 m). The difference is small in the plot, but the comment should say what theory predicts. They correctly recognized that T vs `run` without subtracting theory shows nothing.

Team Bl: interaction plot (A only) on both scales, boxplots by operator group and by square/row, **deviation from theory vs length** (a nice plot, showing systematically short periods). No plot in run order (order unknown).

Review Bl: good plots, but the comment does not use what they show: the deviation from theory is systematically negative and concentrated at operator C. The run-order plot is missing.

Summary 4.3: team G complete, team Bl without the run-order plot. Both could have linked the plots to theory better.

Task 5 – Part A: factorial ANOVA in blocks

5.1 Model $T \sim \text{block} + L \times \theta_0$ on T and $\log T$; why the interaction disappears on $\log T$; choice of scale

Why we ask: to understand that interaction depends on the scale (L03, multiplicative model → log), and to choose the scale for a design without interactions (Part B).

Team G – answer:

- On T: interaction $F = 2.36$, $p = 0.097$; on $\log T$: $F = 0.25$, $p = 0.905$. Length and angle highly significant on both scales. Block not significant ($p = 0.20$ and 0.43).
- Explanation: $T = T_0 f(\theta_0)$, on T the angle adds the same percentage, hence more seconds for a long pendulum; the logarithm turns the product into a sum.
- Why the interaction on T was not detected: its theoretical size is only about 0.03 s; with $\hat{\sigma} = 0.013$ s and 16 df the power is low; non-significance does not mean absence.
- They prefer $\log T$ (additive, physical meaning of coefficients).

Review: excellent, exactly as in L03 and the assignment. Explaining the non-significant interaction through power is exactly what we wanted to hear.

Team Bl – answer:

- On T: block $F = 8.17$, $p = 2 \cdot 10^{-5}$; angle $p = 7 \cdot 10^{-4}$; interaction $p = 0.51$. On $\log T$: interaction $p = 0.46$.
- Explanation $\log T = \log f(L) + \log h(\theta)$; they prefer the log scale. "Our interaction p-values do not establish an interaction, but cannot prove its absence."

Review:

- Correct, good explanation of the scale.
- They do not explain why the interaction that theory predicts on T was not found (with $\hat{\sigma} = 0.054$ s the power for a 0.03 s effect is negligible).
- The significance of blocks ($F = 8.2$) is largely due to the errors in block 3, not to real differences between operators (Task 5.3).

Follow-up question Bl: How large is the theoretical interaction in seconds, and did you have a chance to detect it?

Summary 5.1: both correct. Team G also argues correctly via power.

5.2 Tukey with MS_E and df_E of the model; angles at each length if the effect depends on L; comparison with theory

Why we ask: `pairwise_tukeyhsd()` fits a one-way model with the wrong MS_E (L02/L03). Comparison with theory is the main test of the result.

Team G – answer:

- $\log T$, $n = 9$, HSD = 0.0113: 30−5 +0.88 % (theory 1.69 %), not significant; 45−5 +3.3 % (3.94 %), 45−30 +2.4 % (2.21 %), both significant. All theoretical values inside the simultaneous intervals.
- On T by length with MS_E of the full model: 45−5 significant at all lengths, differences grow with L as theory predicts.
- 30−5 is half of the theoretical value; explained by amplitude decay during the 10 oscillations.

Review: correct (helper `tukey_from_model`, correct MS_E and df_E). The comparison with theory is explicit and well interpreted ("not significant ≠ no effect").

Team Bl – answer:

- Own Tukey via model contrasts (equivalent and correct: half-width $q \cdot SE/\sqrt{2}$).
- $\log T$ marginally: 30−10 0.0178 (theory 0.0154), not significant; 50−10 0.0456 (0.0466), $p < 0.001$; 50−30 0.0278 (0.0312), $p = 0.049$.
- By length on $\log T$ with a theory column: at 0.35 m, 30−10 is even negative (−0.96 %).

Review:

- Technically correct, and the agreement of the marginal differences with theory is **very good** (better than team G).
- A verbal comparison with theory is missing; the "theory" column appears only in the per-length tables.
- The negative 30−10 difference at 0.35 m is not commented; it shows how large the noise is.

Follow-up question Bl: Does your marginal 50−10 difference agree with theory? Why is 30−10 at 0.35 m negative?

Summary 5.2: both correct. Team G interprets better; team Bl is numerically closer to theory (thanks to the wider range of angles).

5.3 Was blocking useful? Relative efficiency of the RCBD vs a CRD

Why we ask: L03 (Montgomery): $\hat{\sigma}^2_{CR} = \frac{(b-1)MS_B + b(a-1)MS_E}{ab-1}$, $R = \hat{\sigma}^2_{CR}/MS_E$ (optionally with a df correction). The plain ratio of MS_E of the models without and with blocks is a different (incorrect) quantity.

Team G – answer: Montgomery's formula + df correction: $R = 1.05$ (T), 0.98 ($\log T$); blocking did not help, on $\log T$ we paid 2 df for nothing. With the caveat that, because of restricted randomization, this is only a rough guide.

Review: correct according to L03, with careful interpretation. A model answer.

Team Bl – answer: $MS_E(\mathrm{CRD})/MS_E(\mathrm{RCBD}) = 1.80$, slightly less with the df correction; blocking improved precision.

Review:

- The formula **does not match L03**. Montgomery's formula gives $R = 1.68$ (recomputed). The conclusion ("blocking helped") is the same.
- Interpretation: the large block effect is mainly block 3 (counting errors) and operator C. Without block 3 the block test is still significant ($p = 0.02$) but weaker. "Blocking helped" here means rather "the block absorbed gross errors" than that operators had stable different habits.

Follow-up question Bl: What formula for relative efficiency was in L03, and why does it differ from the plain ratio of MS_E?

Summary 5.3: team G correct. Team Bl has the right conclusion with the wrong formula.

Task 6 – Part B: Graeco-Latin squares

6.1 Factor model for two squares with ANOVA table and df; the same model for one square

Why we ask: to check understanding of the square structure, rows within squares and saturation (L04).

Team G – answer:

- Check of the square properties, ANOVA with df breakdown (1 + 4 + 2 + 2 + 2 + 6 = 17). Length $F = 7396$, angle $F = 23.3$, $p = 0.0015$, mass $F = 3.93$, $p = 0.081$, rows $p = 0.12$. $\hat{\sigma} = 0.7$ %.
- Length estimates +0.300 and +0.517 vs theory 0.309 and 0.525; angle 45° +2.9 % (theory 3.9 %), 30° +1.0 % (1.7 %).
- One square: 9 parameters, 0 df, $R^2 = 1$; they show that `anova_lm(typ=2)` fails and `typ=1` returns a meaningless $F = 0$.

Review: excellent. Showing the software behaviour for a saturated model goes beyond the requirements.

Team Bl – answer:

- ANOVA on $\log T$: length $F = 672$; angle $F = 3.81$, $p = 0.085$; mass $p = 0.35$; rows $p = 0.14$.
- One square: rank 9, 0 df, "a perfect fit is not evidence of a perfect physical model".

Review:

- Correct.
- **They do not comment that the angle is not significant in Part B** ($p = 0.085$), although it is in Part A. Reason: $\hat{\sigma}$ in Part B is about 2.4 % (recomputed), three times that of team G, with only 6 df. The 50−10 estimate is 0.039 vs theory 0.047.

Follow-up question Bl: Why is the angle not significant in Part B when it is in Part A?

Summary 6.1: both correct. Team G deeper and with comparison to theory.

6.2 The same model on T; comparison of the F-tests for mass and operator on both scales

Why we ask: on the T scale the $L \times \theta$ interaction is aliased with mass and operator, so their tests on T are biased.

Team G: table for both scales; F slightly larger on $\log T$, same decisions; "those tests are the wrong ones" (meaning on T). **Review G:** correct; the wording "those tests" is unclear. They could have added that the bias is negligible at their precision (see 6.3: spurious mass SS $1.7 \cdot 10^{-6}$).

Team BI: table of F and p for mass and rows on both scales. Only a general verbal comparison ("mass and row conclusions still depend on this assumption"). **Review BI:** formally done; interpretation of the difference missing.

Summary 6.2: both done; neither quantified the bias explicitly (team G indirectly in 6.3).

6.3 Aliasing in a 3×3 square: what it means for $L \times \theta_0$ and for conclusions about mass and operators on the T scale; why $\log T$ is the right scale

Why we ask: the key property of a Graeco-Latin square (L04): the interaction of two factors is completely confounded with the main effects of the other two.

Team G – answer:

- Verbally: the 8 df of a square are used up; the interaction (4 df) must lie in the space of rows and mass.
- **Numerically:** projection of the interaction space onto the row + mass space (rank 4, residual $3 \cdot 10^{-15}$).
- Noise-free theoretical periods fitted by the model: spurious mass and operator effects appear on T and vanish to machine precision on $\log T$.

Review: outstanding, the best part of both projects. The only thing missing: aliasing also concerns other interactions, e.g. mass × angle (a lighter bob decays faster, so the angle effect depends on mass). That one is aliased with length and operator, and the logarithm does not remove it.

Team BI – answer: verbally: the four length × angle interaction terms coincide exactly with the four mass and row terms; an interaction can look like a mass or operator effect; a second square does not estimate the interactions; the log scale is supported by theory but does not remove damping or systematic errors.

Review: correct and concise. The remark that the log does not remove damping is exactly the point team G missed.

Follow-up question (both): Which other interactions can the square hide, and does the logarithm remove them? (Mass × angle due to damping.)

Summary 6.3: both correct; team G exceptionally thorough.

Task 7 – Numeric model, the exponent of L and the estimate of g

7.1 Regression of $\log T$ on $\log L$, $x(\theta_0)$, m, operator and block/square; tests $\beta_1 = 1/2$, $\beta_2 = 1$; why $x(\theta_0)$ instead of θ_0^2

Why we ask: moving from the factor model to the physical model (regression in ANOVA, L02); testing against theoretical values, not against zero.

Team G – answer:

- 45 runs, operator + session (A, B1, B2), both in Sum coding, $R^2 = 0.9985$, $\hat{\sigma} = 0.9$ %.
- $\hat{\beta}_1 = 0.5014$ (0.495; 0.508), test of 1/2: $p = 0.65$. Part A alone: 0.508, $p = 0.056$.
- $\hat{\beta}_2 = 0.812$ (0.636; 0.987), test of 1: $p = 0.036$, rejected. Explained by amplitude decay.
- $x(\theta_0)$ vs θ_0^2: SS_E 0.00312 vs 0.00313, table of the difference (0.09 % at 50°), explanation why $\beta_2 = 1$ is a physical hypothesis only with x.

Review:

- Correct and complete, good coding.
- The angle result agrees with the physics of a real pendulum: decay reduces the mean amplitude during the timing, so the effect is smaller than for the release angle.
- The CI for β_1 is probably too narrow because of the unreplicated length setting (Task 3.1). The borderline $p = 0.056$ from Part A points the same way.

Follow-up question G: How would you verify or account for the amplitude decay (e.g. measure the amplitude at the end)?

Team Bl – answer:

- 72 runs, 12 strata (6 blocks of A, 6 rows of B) in Sum coding, mass centred only in B (zero in A, absorbed by the strata).
- $\hat{\beta}_1 = 0.489$ (0.472; 0.506), $p = 0.19$; $\hat{\beta}_2 = 0.93$ (0.55; 1.31), $p = 0.71$ (p recomputed).
- x vs θ_0^2: only the claim "describes 50° better", no demonstration.

Review:

- A well-thought-out model (handling the fixed mass in A is elegant and correct).
- "β_2 compatible with 1" is true, but the interval (0.55; 1.31) is so wide that the test says almost nothing. This should be stated.
- The argument for x vs θ_0^2 is missing.

Follow-up question Bl: What is the smallest β_2 your experiment would still not reject? Why is $x(\theta)$ better than θ^2?

Summary 7.1: both have a length exponent consistent with 1/2. Team G estimated the angle precisely and found a physically explainable deviation; team Bl's angle agrees with theory only because of imprecision.

7.2 Numeric model on one square: residual df compared with the factor model

Why we ask: a numeric factor costs 1 df instead of 2, so even one square allows a test.

Team G: 9 runs, 6 parameters, 3 df (factor model 0 df), coefficients shown; "each numeric factor costs one df instead of two". **Correct. Team Bl:** rank 6, 3 df; "the gain comes from assuming a particular response shape, not from extra data". **Correct**, with a good remark about the price. **Summary 7.2:** both correct.

7.3 Estimate of g with fixed $\beta_1 = 1/2$, $\beta_2 = 1$; CI by transformation; coding; cause if 9.81 is outside

Why we ask: transforming the interval, the meaning of the intercept under the coding, and above all critical thinking about systematic errors.

Team G – answer:

- $\hat{g} = 9.84$ (9.78; 9.91), 9.81 inside. Part A alone 9.92 (9.85; 10.00), just outside; Part B 9.80 (9.71; 9.90).
- Demonstration that treatment coding would give an intercept corresponding to 10.03 (a different quantity).
- g by length in Part A: 10.04 / 9.86 / 9.88. Largest for the shortest pendulum → additive error in L (0.3–0.8 cm) or in t_N (−0.04 to −0.14 s); distinguished by the $1/L$ vs $1/\sqrt{L}$ dependence.

Review:

- Excellent, exactly what the task asked for.
- A check by operator (recomputed) would add: A 9.78, B 9.94, C 9.91, so part of the deviation is due to timing.
- The intervals do not include the uncertainty of the length; the team knows this.

Team Bl – answer:

- $\hat{g} = 10.25$ (10.08; 10.43), 9.81 outside. Part A 10.28, Part B 10.22 (recomputed).
- The interval describes only statistical uncertainty, not systematic errors (well said).
- Possible causes listed in general (length, counting, timing, N, angle, damping); "these data cannot identify the cause".

Review:

- Transformation and coding correct; the careful interpretation of the interval is good.
- **"The data cannot identify the cause" is not true; the data largely can:**
  - g by length in Part A increases: 10.14 / 10.29 / 10.41. A length error would give the largest deviation at the shortest pendulum, i.e. the opposite pattern. It is not the length.
  - g by operator: A 9.85, B 10.07, **C 10.91** (recomputed). The ratio T/theory for operator C is 0.942 in Part A and 0.917 in block 3.
  - Without block 3, Part A gives $\hat{g} = 10.02$.
  - The main cause is therefore timing/counting by operators B and C, especially block 3. Operator A measured periods consistent with theory (ratio 0.996–1.005).

Follow-up question Bl: Compute g separately for each operator and each length. Which cause can you exclude?

Summary 7.3: team G answered in an exemplary way and its g agrees. Team Bl's g is 4.5 % too high, and the team did not analyse the cause that its own data reveal.

Task 8 – Does mass matter? Equivalence, not significance

8.1 Mass effect with a 95 % CI from the factor model (3 − 1 ring) and the numeric model

Why we ask: an effect should be described by its size and an interval, not only by a p-value.

Team G – answer: factor model +1.02 % (−0.02; +2.08), numeric +1.02 % (−0.06; +2.11). The interval contains 0, not rejected.

Review:

- Calculation correct.
- Physical context is missing, though interesting: the positive sign is consistent with damping. A lighter bob decays faster, has a smaller mean amplitude and a shorter period. In the data the 1-ring runs at 45° are 2.0 % and 1.1 % below theory, the 3-ring runs −0.7 % and +0.4 % (recomputed). That would be a mass × angle interaction, which is aliased in the square.
- At the same time, mass is confounded with run order (Task 3.2), so +1 % may also be drift.

Follow-up question G: Does the +1 % effect have a physical explanation? How would you distinguish it from drift?

Team Bl – answer: factor model −2.10 % (−5.43; +1.35), numeric −2.09 % (−5.56; +1.50).

Review: correct. The interval is very wide because of the large $\hat{\sigma}$ in Part B. The negative sign is opposite to what damping would predict, but within the noise.

Summary 8.1: both computed correctly. Team G has a precise estimate, but confounded with time; team Bl's estimate is imprecise.

8.2 Choice and justification of Δ; TOST, i.e. 90 % CI inside $(-\Delta, \Delta)$

Why we ask: an equivalence test is the right answer to "does mass have no effect?" (L01: power, type II error). The margin is chosen **in advance** from a practical requirement.

Team G – answer:

- They tried $\Delta = 1$ % and 2 %, justified by $dg/g = 2\,dT/T$ (i.e. g to 2 % or 4 %).
- 90 % CI (+0.19; +1.86) %: equivalence not shown at 1 %, shown at 2 %.

Review:

- The TOST method via the 90 % CI is correct.
- **Margin chosen after looking at the data**: they picked the one that passes. The assignment suggested g to 1 %, i.e. $\Delta = 0.5$ %; that one fails.
- 2 % of the period = 4 % in g ≈ 0.4 m/s², six times the width of their own CI for g. Calling it a "highly precise margin" is an overstatement.
- **The 90 % CI does not contain zero.** At the 5 % level (one-sided) the data rather indicate a positive effect. The team does not comment on this.

Follow-up questions G: When is the margin chosen, and why? What does it mean that the 90 % CI excludes zero?

Team BI – answer: $\Delta = 0.5$ % justified by g to 1 %, admitted to be chosen after the measurements. A full TOST with t and p (TOST $p = 0.85$ and 0.81): equivalence not shown.

Review: correct and honest. A reasonable, single margin. They could have added how many runs would be needed to show equivalence with their $\hat{\sigma}$.

Summary 8.2: team BI correct. Team G has the right procedure but bad practice (margin chosen by the result) and an overstated conclusion.

8.3 The difference between "the mass effect is not significant" and "the mass effect is negligible"

Why we ask: this is the point of the whole task and of the remark in the assignment about last year's teams.

Team G: correct explanation (failing to reject H_0 ≠ proof of zero; equivalence = the whole 90 % CI inside the margin). The wording "it is guaranteed to be too small to matter" is an overstatement (statistics guarantees nothing). **Team BI:** brief and correct: "Failure of either equivalence test does not prove mass matters; it means a practically relevant effect cannot be ruled out."

Summary 8.3: both understand the concepts. Team G overstated the wording and, more importantly, the practice (8.2).

Task 9 – Model diagnostics

Question: for the models of Parts A and B on the chosen scale: residuals vs fitted values, normal Q-Q plot, residuals vs each factor, operator and **run order**; comments and proposals. **Why we ask:** checking the ANOVA assumptions (L02), above all independence and time trends.

Team G – answer:

- Part A: residuals in three groups by length; Q-Q approximately normal, two points in the upper tail at 0.35 m; **heteroscedasticity**: about three times larger spread at 0.35 m, explained by a constant timing error, relatively larger for a short period. Run order: no trend, largest residuals in runs 17–23, when 0.35 m was measured.
- Part B: no problems, no trend in run order, smaller residuals (with caution because of the number of parameters).

Review:

- Complete and well interpreted, including the physical explanation of the heteroscedasticity.
- They did not realize that in Part B **the residuals-vs-run-order plot cannot show drift**: the order within a square is the same as mass, so any drift would be absorbed entirely into the mass effect estimate and

would not appear in the residuals. Similarly in Part A: the cluster of residuals in runs 17–23 cannot be separated from the length effect.
- A remedy for the heteroscedasticity (larger N for the short pendulum, weighted regression) only appears in Task 11 ($N = 20$).

Follow-up question G: Could the residuals-vs-run-order plot in Part B reveal drift when the order is identical with mass?

Team BI – answer:

- Six panels (no run order), Cook's distance table, Shapiro–Wilk (A: $p = 0.034$, B: $p = 0.51$).
- Comment: group C in repetition 1 has short periods; a block removes an average shift, not a shift that depends on length; Part B has high leverage, so small residuals prove nothing; these plots cannot verify independence.

Review:

- Statistically a very good commentary (leverage, non-additive block effect, caution with Shapiro).
- **Residuals vs run order are missing** (order unknown); requirement not met.
- The detected outliers (Cook's D 0.28 for $t_5 = 6.98$ s) led to no action: neither a check of the protocol nor a sensitivity analysis without them.

Follow-up question BI: How do the conclusions change without block 3? (Answer: $\hat{\sigma}$ 0.054 → 0.044 s, $\hat{g}_A$ 10.28 → 10.02.)

Summary 9: team G complete, but overestimated what the run-order plot can show. Team BI has a deeper statistical commentary, but no run-order plot and no sensitivity analysis.

Task 10 – Comparison of the two designs

10.1 SE of the angle (50° − 10°) and length effects on $\log T$ in Parts A and B; per-run comparison (variance × number of runs)

Why we ask: to understand that both balanced designs have the same per-run efficiency for a main effect: $\mathrm{var}(\hat{\tau}_i - \hat{\tau}_j) \cdot n = 2\sigma^2/(n/3) \cdot n = 6\sigma^2$. The difference comes only from σ, not from the design.

Team G – answer: SEs taken from the coefficients `C(theta_deg)[T.45]` and `C(L_m)[…]` of the model **with interaction**: A 0.0076, B 0.0043; per-run ratio B/A = 0.21. Conclusion: "Graeco-Latin square is definitely more efficient … blocking is applied to both operator and mass while in the latter only to operator."

Review (error):

- In the model `C(L_m)*C(theta_deg)` with treatment coding, `C(theta_deg)[T.45]` is the **simple effect at** $L = 0.35$ ($n = 3$ per cell), not the main effect. Its SE is $\sqrt{3}$ times larger. The correct SE of the main effect in A is $\sqrt{2 \cdot 8.67 \cdot 10^{-5}/9} = 0.0044$.
- The correct per-run ratio is just the ratio of the MS_E: $0.0074^2/0.0093^2 \approx 0.63$, and that is not significant (F-test, $p \approx 0.6$).
- The conclusion is **conceptually wrong**: mass is a factor of interest in the square, not a block; the square is not "more efficient per run", it just estimates more factors from the same number of runs.
- The identical SEs for angle and length should have been a warning sign (both are the SE of a single cell).

Follow-up questions G: What does the coefficient `C(theta_deg)[T.45]` estimate in a model with interaction? What is the per-run variance of a main effect in the two designs?

Team BI – answer: marginal contrasts via the averaged design matrix: A SE 0.0114, B 0.0141; var·n 0.00701 (A) and 0.00360 (B). The difference is attributed to the different N (5 vs 10) and noise, not to the design.

Review: correct. The numbers equal $6\,MS_E$ in both parts. They could have stated the general relation var·n = $6\sigma^2$ explicitly.

Summary 10.1: only team Bl correct. Team G picked the wrong parameter and drew a wrong conclusion from it.

10.2 What can A estimate that B cannot and vice versa; assumptions of each design and whether the data support them

Why we ask: the price and the gain of a Graeco-Latin square (research question 4).

Team G – answer: A estimates the $L \times \theta$ interaction, not mass; B four main effects from 18 runs, no interactions ("not enough df"). Assumption of A: block × treatment additivity, "supported by RE ≈ 1". Assumption of B: no interactions; violated on T, hence $\log T$.

Review:

- The split of capabilities is correct (missing: A also estimates operators as blocks).
- "Not enough df" is not the precise reason; the reason is aliasing (Task 6.3, which they demonstrated so well themselves).
- "RE ≈ 1 confirms block × treatment additivity" is a **non sequitur**. RE measures the size of the block effect, not the interaction of block and treatment. That would be shown by residuals by block and treatment or by Tukey's test for non-additivity (L03).
- The B assumption on $\log T$ is supported by theory, but a possible mass × angle interaction (damping) remains (see 8.1).

Team Bl – answer: a table: A length × angle with interaction, mass fixed; B mass, no interactions; limitation of A "setting-dependent block effects enter the error estimate", of B "interactions can be mistaken for separate factor effects"; the log scale is supported by theory but cannot be verified.

Review: correct and precise, including where a violated assumption ends up in each design.

Summary 10.2: team Bl better. Team G has a logical error in the additivity argument.

10.3 Only 18 runs for research questions 1–3: which design and why

Why we ask: synthesis: question 3 (mass) needs mass as a factor, which the RCBD in Part A does not have.

Team G: Graeco-Latin squares, "the superior and only viable choice", because they estimate four effects and "block on operators and mass". **Review G:** a defensible choice, but the justification repeats the "blocking on mass" error, and "only viable" is an overstatement (e.g. an 18-run factorial with mass and a numeric model would also work).

Team Bl: two independently randomized squares plus the numeric model as a complement; interaction checks are given up, and even this may not suffice for ±0.5 % mass equivalence; if interactions were the priority, a replicated full factorial. **Review Bl:** an excellent, balanced answer.

Summary 10.3: both chose squares; team Bl justified it better.

Task 11 – Reflection and conclusions

11.1 Actual power for 10° vs 30° with $\hat{\sigma}$ from the main experiment; did the pilot estimate of σ hold?

Why we ask: a retrospective check of the planning (L01), lessons for next time.

Team G: on $\log T$: $D = x(30) - x(10)$, $n = 9$, $df = 16$ → **82 %** (planned 99.7 %). The pilot underestimated σ (0.58 % vs 0.93 %) because it used one operator and one length; the Part A value lies in the pilot's 95 % CI (0.0066; 0.0175) s. 30° vs 5° still was not detected because the observed effect was half of the theory. **Review G:** exemplary, including the interval for σ and the explanation.

Team Bl: $\hat{\sigma}_A = 0.0537$ s vs pilot 0.0290 s (1.85×); power with $b = 6$ is **21 %**. They explicitly reject "observed power" from the measured effect – correct. **Review Bl:** correct and well informed. The explanation why σ is larger ($N = 5$, block 3), which they have elsewhere, is missing here.

Summary 11.1: both correct. Team G had a good plan that worked; team Bl had an underpowered design and correctly admitted it.

11.2 Answers to the four research questions

Team G:

- RQ1: $\beta_1 = 0.501$ consistent with 1/2; $g = 9.84$ (9.78; 9.91), 9.81 inside. They do not mention that Part A alone excludes 9.81 (that is in Task 7).
- RQ2: angle clearly detected, $\beta_2 = 0.81$ smaller than 1, explained by damping. Good.
- RQ3: "We showed that mass has no practically relevant effect" at a 2 % margin, "highly precise". **Overstated** (Task 8.2), and confounded with time.
- RQ4: "extreme efficiency per run by blocking on … Operator and Mass". **Wrong** (Task 10.1).

Team Bl:

- RQ1: exponent 0.489, $g = 10.25$ (10.08; 10.43) "conditional on timing and calibration assumptions". They do not say explicitly that 9.81 is outside.
- RQ2: 50° − 10° in Part A +4.67 % vs theory +4.77 % (recomputed), very good agreement.
- RQ3: equivalence not shown; a non-significant test does not establish irrelevance. Correct.
- RQ4: squares study more factors with fewer runs at the price of additivity and 6 df; the factorial estimates the interaction but not mass. Correct.

Summary 11.2: team Bl answers more carefully and more correctly (RQ3, RQ4). Team G has better results (RQ1, RQ2) but overstated RQ3 and got RQ4 wrong.

11.3 What to change in a repeat with the same time budget

Team G: $N = 20$, a pilot with all operators and the shortest length, maximum angle 50–60°, measure L to the centre of the rings. Concrete and correct. **Missing** the main point: how to randomize more even with one pendulum (random order of lengths, reset the length in each block, randomize angles and operators within a length) and a truly random order in Part B.

Team Bl: the same N everywhere and a pilot with the same procedure, video or a photogate, careful measurement of L, the same release procedure, record anomalies, plan the number of repeats from the precision needed for mass. Correct but more general. **Missing**: acknowledge the restricted randomization, record and randomize the run order, and check the number of oscillations (block 3).

Summary 11.3: both proposed sensible improvements of the measurement; neither named the main shortcoming of the design (randomization and run order).

Task 12 – Submission and reproducibility

Why: the notebook must run in Colab from top to bottom and read the data from a URL.

Team G: reads from the raw URL of the fork (`paletvoj`) with a local fallback; saved with outputs; the CSV has the required columns (`date_time` empty). Minor points: `if "data" not in globals()` (non-standard); an f-string with nested double quotes in Task 5c works only in Python ≥ 3.12 (Colab is fine).

Team Bl: first the local CSV, then the fork URL (`CasualMathEnjoyer`); outputs saved; the CSV has extra columns `source_order`, `group`, `code` (fine); `run` and `date_time` empty. Conclusions are printed via `display(Markdown)`, so the numbers are not visible in a text view of the notebook; they render on GitHub. The assignment text and the Colab badge were removed (does not matter).

Summary 12: both reproducible. Team Bl did not provide the required CSV content (`run`).

Overall summary by task

Task	Team G	Team Bl	What to improve (if neither did it)
1.1 apparatus, N	good; one pendulum (allowed), restriction acknowledged	weaker description; one pendulum not acknowledged; $N = 5$	record measured L and ring radius
1.2 nuisance factors	**model answer**	no randomization	–
1.3 A vs B	correct	missing	–
2.1 pilot	correct (mixed pilot)	different N, trend and −3.8 % not noticed	–
2.2 power, b	**correct, plan from the pilot**	formula correct, b not planned	plan with the upper limit of σ
2.3 df of squares	correct	correct	–
2.4 plan in advance	done	not done	–
3.1 run sheet A	restricted randomization, acknowledged, order recorded	not acknowledged, post hoc plan, order unknown	randomize what is possible (order of lengths, angles within a length)
3.2 run sheet B	**searched seeds**, mass = time	post hoc plan	**randomize in advance**
3.3 `run`	yes	no	–
4 data and plots	complete	better structural checks, no run order, block 3 not investigated	investigate suspicious runs
5.1 ANOVA, scale	**excellent**	correct	–
5.2 Tukey, theory	correct, good interpretation	correct, closer to theory	–
5.3 RE	**as in L03**	different formula (1.80 vs 1.68)	–
6.1 squares, one square	excellent	correct, angle in B not commented	–
6.2 T vs $\log T$	correct	formal	quantify the bias
6.3 aliasing	**outstanding, numerical**	correct, verbal	mention mass × angle (damping)
7.1 regression	correct, $\beta_2 = 0.81$ explained	correct, weak test of β_2	–
7.2 one square numeric	correct	correct	–
7.3 g	**exemplary**, 9.81 in CI	g outside, cause not analysed	analysis by operator
8.1 CI of mass	correct	correct	–
8.2 TOST	margin after the data, overstated	**correct**	–
8.3 not significant vs negligible	correct, overstated wording	correct	–
9 diagnostics	complete	no run order, no sensitivity analysis	a run-order plot cannot show drift confounded with a factor
10.1 per run	**error** (simple effect)	**correct**	–
10.2 assumptions	non sequitur about RE	correct	–
10.3 18 runs	defensible, wrong justification	excellent	–
11.1 actual power	exemplary	correct	–
11.2 research questions	RQ3 overstated, RQ4 wrong	careful and correct	–
11.3 improvements	concrete	more general	both: randomization and recording the order

Team G ran the experiment more carefully (pilot → power → plan, `run`, notes), reached a precision of 0.9 % and results consistent with theory (exponent, g, angle with a physically explainable deviation). The analysis is in places exceptional (aliasing, g by length, RE). They acknowledged the restricted randomization with one pendulum and

recorded the order. Weaknesses: in Part B they claim a randomization that the searched seeds contradict; consequences of the unreplicated length setting not thought through; the error in Task 10 and overstated conclusions about mass.

Team BI has statistically more careful and more correct interpretations (comparison of designs, TOST, assumptions). The weaknesses are in the execution: $N = 5$, plan made after the measurements, unacknowledged restricted randomization and missing run order, uninvestigated counting errors in block 3, and g 4.5 % off without analysing the cause that its own data reveal (operator C).